



P18

A Broad-then-Deep Two-Stage Memory Flywheel: Submodular Active Sampling of BRS and DRS

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A Broad-then-Deep Two-Stage Memory Flywheel: Submodular Active Sampling of BRS and DRS

Working Title (EN): A Broad-then-Deep Memory Flywheel: From Coverage-Aware Sampling to Submodular Depth-Probing, Extending UDOS P4 with the RSI-Agent Lesson

In-volume engineering paper P18 • upgrading v7.5.0 P4 "Coverage-aware data flywheel" to a "breadth scan + depth probe" two-stage structure • connecting to P6 identifiability blind zone • UDOS Reasoning Engine v7.6.0 • Fang Wenxin • 2026-09-21

One sentence first: UDOS's current data flywheel is "collecting data to remove blind zones," but it only asks "where not yet seen," not "where most incomprehensible." RSI-Agent (arXiv:2609.15364) with the "broad-then-deep" strategy proves: **first using breadth memory to survey the environment, then throwing compute at hard cases and hidden constraints, memory compound interest only then rolls up.** This paper lands this idea as UDOS's two-stage flywheel specification—BRS breadth scan covering the state space, DRS depth probe focusing on identifiability blind zones, and clearly marks: this is a **design proposal** • **to implement**, not measured benefits already produced.

Evidence caliber (stated only once in the whole paper): this paper is an architecture design paper, **containing no new UDOS measured numbers.** The cited external paper is RSI-Agent (arXiv:2609.15364, existence verified online 2026-09-21), its "Kimi-K3/GLM-5.3 surpass GPT-6-class closed-source models on designated benchmarks" is paper/media report, not independently reviewed; this paper does not take that performance conclusion as evidence for the UDOS scheme holding, only borrowing its "broad-then-deep" strategy structure. UDOS old numbers all point back to v7.5.0 papers and `udos-engine/reports7/*.json`, evidence grades marked in text (verified / cpu-proto / cpu-proxy / unverified). The two-stage flywheel this paper proposes is a **design proposal** • **to implement**, wherever expected benefits lack experiment support all marked [RESULT NEEDED].

Abstract

Data collection is the fuel for Agent system self-improvement. UDOS v7.5.0's P4 paper already upgraded "passively collecting logs" to "coverage-aware active collection": the system measures its own coverage gap on the state space, prioritizing collecting tasks in regions never reached. This step is correct, but it only solved the "breadth" problem. This paper points out the missing second stage: **regions reached do not equal regions understood.** P6's identifiability empirical study of hidden dynamics parameters gives an honest negative result—the acceleration scalar skill estimate is negative (about -0.508 , single-seed pilot, evidence grade cpu-proxy, pointing back to P6, this paper not recomputing), meaning the system in that direction "sampled, yet did not recognize the truth." RSI-Agent (arXiv:2609.15364, Sibozhu et al.)'s approach precisely fills this gap: it coordinates three Agent types—curriculum (problem setter), actor (solver), verifier—first using broad recursive self-exploration (BRS) to spread breadth memory in diverse environments, then deep recursive self-exploration (DRS) focusing on hard cases, hidden constraints and boundary conditions. This paper translates this "broad-then-deep" into UDOS engineering specifications: the BRS stage targets coverage maximization, using deterministic submodular functions (e.g. facility-location) to select points; the DRS stage targets **identifiability information gain**, concentrating the sampling budget on blind zones "where the prediction error structure along parameter directions is most indistinguishable." The two stages share the same memory

body, forming a "breadth memory + depth memory" layering. This paper also gives the design's failure boundaries: memory is environment-specific, RSI-Agent's capability improvement bound to concrete digital environments, cannot transfer across environments (the reason this paper stays restrained when citing it); BRS+DRS is only a data collection strategy, not changing the quality saturation law P9 established that "isomorphic copies bring no independent information." This paper is a design specification, effects pending [RESULT NEEDED] .

Keywords: data flywheel; active learning; submodular function; identifiability; recursive self-improvement; memory layering; UDOS P4; UDOS P6

Structured Abstract (background problem → method/argument → evidence → contribution)

- **Background problem:** P4's coverage-aware flywheel answered "where not yet visited," but did not answer "where visited yet not understood." P6 exposes a more hidden blind zone class—sampling density sufficient, truth still not identifiable. Only expanding breadth, such blind zones are repeatedly passed yet never close.
 - **Core argument:** active collection should be two stages. BRS (broad recursive exploration) targets state-space coverage maximization, using submodular point selection; DRS (deep recursive exploration) targets **identifiability information gain along single-parameter directions**, pressing budget to the non-identifiable blind zones P6 identified. The two stages share the memory body, first broad then deep, alternating iteration.
 - **Evidence clues:** (1) RSI-Agent's BRS/DRS two-stage strategy structure (arXiv:2609.15364, paper report, existence verified); (2) P6's identifiability negative result (acceleration scalar skill negative, cpu-proxy, single-seed pilot); (3) P4's existing coverage-aware collection yield improvement (pointing back to P4, evidence grade per that paper); (4) P9's quality saturation law ($mse(k)=a+b/k$, cpu-proto) as the "breadth $\neq$ quality" constraint.
 - **Contribution:** expanding P4 from single-stage to a two-stage flywheel specification; giving BRS/DRS respective objective functions and switching criteria; clarifying DRS's "depth" should anchor on identifiability blind zones rather than mere difficulty; drawing three failure boundaries of this design.
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1 Problem: Covered Does Not Equal Understood

1.1 The Half P4 Already Solved

v7.5.0's P4 paper did one correct thing: changing the data for training Agents from "runtime byproduct" to "actively planned product." The system first estimates its own coverage distribution on the state space, then prioritizes collecting regions never reached, or reached far fewer times than the mean. This is conceptually equivalent to changing "collecting data" from "waiting for it to happen" to "going where it never happened." P4 reports active collection yield improved relative to passive collection (pointing back to P4, concrete values per that paper's `reports7/*.json`, this paper not restating, not newly creating).

This stage's objective function is **coverage rate**: the proportion of cells in the state space "touched." Its implicit assumption is—as long as touched, learned.

1.2 The Other Half P6 Exposes: Touched $\neq$ Understood

P6 paper did a more uncomfortable experiment: it does not look at "how much collected" but "whether truth parameters can be uniquely recovered from collected data." The conclusion is layered—on certain parameter directions, even with equal sampling and coverage looking not bad, parameters not identifiable; among them the acceleration scalar skill estimate is even negative (about -0.508 , single-seed pilot, cpu-proxy). This is not "sampling not enough" but **the current observation structure is insensitive to that direction**: the system repeatedly passes this direction, yet no single observation can distinguish it from its neighboring directions.

In a metaphor: BRS solves "which map block not yet visited"; P6 reveals "some places you pass daily, but because the angle is too poor, the photos taken cannot recognize what it is."

A breadth-only flywheel is ineffective for such blind zones—it will walk the blind-zone cells again, coverage count +1, identifiability unmoved. This is the second stage this paper fills.

Figure 1 Breadth flywheel vs breadth-depth two-stage flywheel

Figure 1 Left: single-stage coverage flywheel, only counting "visited," blind zones repeatedly passed yet not closed; right: BRS breadth coverage + DRS depth probe, DRS pressing budget to identifiability blind zones. Conceptual illustration, not measured data.

1.3 Why Fill Now: RSIAgent's Structure Is Borrowable

RSIAgent (arXiv:2609.15364, Sibo Zhu et al., 49 pages, 2026-09) is a **training-agnostic** multi-Agent self-improvement framework: it does not change model weights, only memory. It coordinates three roles—curriculum (problem-setting Agent), actor (solving Agent), verifier—and adopts **broad-then-deep** exploration:

- **BRS (broad recursive self-exploration)**: spreading in diverse environments, the goal being to record fully "what the task looks like under what conditions";
- **DRS (deep recursive self-exploration)**: repeatedly digging deep on already-known hard cases, hidden constraints, boundary conditions, the goal being to thicken the memory of "this class I just cannot do right."

The paper reports: under this memory flywheel, open-source Kimi-K3 and GLM-5.3 surpassed GPT-6-class closed-source models on OSWorld-v2 and Agent's Last Exam (paper/media report, not independently reviewed). **This paper does not cite this performance conclusion as the reason the UDOS scheme holds**—it is bound to specific environments and models. This paper borrows only a weaker, more stable thing: **"first broad then deep" is a more reasonable memory collection order than "only broad."**

2 Two-Stage Flywheel Specification (design proposal • to implement)

2.1 State and Memory Notation

Following P4's state space $\mathcal{X}$, recording UDOS task execution traces as $\mathcal{D}_t = \{(x_i, y_i)\}$, x_i states, y_i observable outcomes. The memory body $\mathcal{M}_t$ is structured experience refined from $\mathcal{D}_t$, reusable by later tasks. The two stages share the same $\mathcal{M}_t$.

2.2 BRS Stage: Targeting Coverage Maximization

BRS's objective function follows the submodular view—the standard "diminishing marginal returns" form in active learning. Let the candidate state pool be $\mathcal{C}$, the selected set S , the gain of selecting the next point x using facility-location-type gain:

$$g_{\text{BRS}}(x | S) = \max_{x' \in \mathcal{C}} \text{sim}(x, x') \quad \tag{1}$$

where sim is state similarity (embedding cosine or workspace fingerprint similarity). BRS each step picks the x making the gain drop slowest, i.e. **farthest from the selected set**. This greedy under submodular conditions has a $(1-1/e)$ approximation guarantee (classic result, unrelated to this paper's data). The BRS stage's exit condition is: the new point's marginal contribution to the coverage function below threshold τ_{cov} (design parameter, value pending [RESULT NEEDED] calibration).

Plainly: BRS is "picking things most unlike already seen to view," filling the map.

2.3 DRS Stage: Targeting Identifiability Information Gain

DRS is this paper's key increment relative to P4. It does not pick "farthest" but the parameter direction **"current observation most cannot separate truth."** Let the hidden parameter vector be θ (same source as P6's identifiability analysis), Fisher information approximately characterizing observation's ability to distinguish θ :

$$I(\theta | D) = \sum_{(x,y) \in D} \nabla_{\theta} \log p(y|x, \theta) \nabla_{\theta} \log p(y|x, \theta)^{\top} \quad \tag{2}$$

DRS's point selection goal is **the direction corresponding to minimizing I 's smallest eigenvalue**—i.e. throwing the next sample to the direction "the information matrix is most barren," making that direction identifiable as soon as possible. The blind zone in P6 "acceleration scalar skill estimate negative" is precisely the engineering manifestation of I approaching singular in that direction.

Plainly: DRS does not ask "where not visited" but "which direction I looked a long time yet cannot distinguish true/false," then specifically adding illumination in that direction.

Safety note: equation (2) is **the formalization of the design goal**, not a number UDOS already computed. Fisher information on black-box Agents needs approximation (e.g. empirical Fisher or proxies based on verifier gradients), approximation schemes and numerical calibration both [RESULT NEEDED] .

2.4 How the Two Stages Alternate

The two stages are not a one-time waterfall of "finish BRS then DRS" but **flywheel-style alternation**:

$$M_{t+1} = \text{Merge}(M_t, \text{Sample}_{\text{BRS}}(M_t) + \text{Sample}_{\text{DRS}}(M_t)) \quad \tag{3}$$

Each round's budget allocated proportionally (design initial: BRS 60% / DRS 40%, pending [RESULT NEEDED] tuning). Switching criteria driven by two quantities: coverage marginal ΔCov (whether BRS still has new map) and identifiability marginal $\Delta \lambda_{\min}(I)$ (whether DRS still has unclosed blind zones). When ΔCov bottoms first, breadth is exhausted, should continue pressing budget toward DRS; when $\Delta \lambda_{\min}$ approaches zero, blind zones closed, should return to BRS opening new domains.

Table 1 Contrast of BRS and DRS (design specification)

Dimension	BRS broad recursive exploration	DRS deep recursive exploration
Goal	state-space coverage maximization	single-parameter direction identifiability maximization
Point selection	farthest from selected set (submodular facility-location)	information matrix most barren direction
Borrowed from	P4 coverage-aware collection	P6 identifiability analysis
Exit criterion	coverage marginal $< \tau_{\text{cov}}$	information matrix smallest eigenvalue stabilizing
Risk	collected but not understood	tunneling, overfitting to a few hard cases
Evidence grade	design proposal • to implement	design proposal • to implement

2.5 Difference from RSIAgent (Why Cannot Copy)

RSIAgent's DRS focuses on "hard cases, hidden constraints, boundary conditions"—its "deep" is deep in the **task difficulty** sense. This paper corrects it to deep in the **identifiability** sense: for UDOS, a case "hard" does not equal "identifiable to the model." A simple but perspective-tricky task may contribute more to closing Fisher-barren directions than a complex but isomorphic-to-existing-observations task. This correction comes from P6's lesson—P6's negative result is precisely not "task too hard" but "observation structure insensitive to that direction."

3 Connection: BRS+DRS Landing in UDOS Existing Papers

Figure 2 Position of BRS/DRS in the UDOS volume

Figure 2 BRS carries P4's coverage-aware collection, DRS carries P6's identifiability blind zones; the two share the memory body, constrained by P9's quality saturation law. Conceptual illustration.

3.1 Carrying P4: Expanding Single Stage to Two Stages

P4's coverage-aware collection is this paper BRS's predecessor. The only upgrade point: beyond the coverage criterion, add an identifiability criterion, letting part of the budget (design initial 40%) no longer flow to "new cells" but to "barren directions." This change does not break P4's existing contracts—it only adds a DRS arm in the sampling selector.

3.2 Connecting P6: Turning Negative Results into DRS Targets

P6's identifiability grading (which parameters identifiable, which not, which directions skill estimate abnormal) is DRS point selection's **prior map**. This is this paper's most direct engineering value: P6 is diagnosis ("here not understood"), DRS is prescription ("specifically illuminate here"). P6's acceleration scalar negative result (about -0.508 , cpu-proxy, single seed) is DRS's first formal target.

3.3 Constrained by P9: Breadth Does Not Equal Quality

Must reaffirm P9's quality saturation law ($\text{mse}(k)=a+b/k$, cpu-proto): **newly added Agents if not carrying independent information, scale expansion produces no capability**. New states BRS collects, new cases DRS digs, must all produce **independent information** in the memory body, otherwise only

recording the same experience twice. The two-stage flywheel therefore is not "collect more" but "collect with more discrimination"—fully consistent with P9, not P9's counterexample.

4 Drawbacks / Failure Boundaries / Falsifiable Conditions

4.1 Drawbacks (restraint that must be kept when citing RSIAgent)

1. **Memory is environment-specific.** RSIAgent's capability improvement bound to concrete digital environments like OSWorld-v2, Agent's Last Exam; its title "in New Environments" means "learning in new environments," **not "learned capability transfers across environments."** This paper borrows its two-stage structure, never its "universal self-evolution" implication.
2. **Self-reported performance cannot be directly moved.** "Open source surpasses closed source" is paper/media report, not independently reviewed, and highly dependent on task and model versions, cannot serve as argument for UDOS scheme expected benefits.
3. **DRS has a tunneling risk.** Pressing all budget to identifiability blind zones may make the system overfit on a few extreme cases, while normal task quality not rising but falling—precisely why BRS/DRS ratio needs experimental calibration rather than head-fixed.

4.2 Failure Boundaries

- If the state space's "non-identifiable" is essentially **observation structure missing** (sensors/tools cannot see that direction at all), then DRS no amount of sampling can close—that is a tool-layer problem, not a collection strategy problem, should upgrade to P10 handoff contract or tool selection rather than continuing to add sampling.
- If the memory body $\$M_t\$$ does not support cross-stage reuse (what BRS records DRS cannot read), the two stages degrade into two non-communicating samplers, flywheel stopping. This requires the memory schema shared by both stages, the implicit premise for this paper's specification to hold.

4.3 Falsifiable Conditions

- **F18-a:** if under the same budget, BRS+DRS two stages relative to single-stage BRS, the **identifiability blind zone closing speed** ($\lambda_{\min}(I)$ improvement rate) has no statistical improvement, then this paper's "depth probe" design proposal is falsified, should return to single stage.
- **F18-b:** if after raising the DRS budget proportion, normal tasks' average quality significantly drops (overfitting to hard cases), then the two-stage ratio needs an upper-bound constraint, this paper's "40% DRS initial" direction correct but value needs fixing.
- The above criteria all need ≥ 30 seeds and 95% CI, currently design proposal, [RESULT NEEDED] .

5 Conclusion

P4 answered "visited or not," P6 answered "understood or not," this paper sews the two into one flywheel: BRS responsible for "filling the map," DRS responsible for "illuminating through the not-understood directions." This is the UDOS-ization correction of RSIAgent's "broad-then-deep" strategy—borrowing its structure, not its performance promise; correcting "task-difficulty depth" to "identifiability depth." The two stages share the memory body, alternately driven by coverage marginal and

identifiability marginal, a **design proposal** • **to implement** specification, its holding or not adjudicated by F18-a/b in controlled experiments.

Breadth determines how much world you have seen, depth determines how much of the seen world you understand. Only broad not deep, blind zones will be repeatedly passed by you, forever passed.

References

Classic active learning and submodular

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2. MacKay, D. J. C. (1992). Information-Based Objective Functions for Active Data Selection. Neural Computation, 4(4), 590–604. (source of information gain/Fisher active sampling)

2026 frontier material (existence verified online 2026-09-21; performance numbers paper/media reports, not independently reviewed)

1. Zhu, S., et al. (2026). RSIAgent: Autonomous Exploration for Recursive Self-improvement in New Environments. arXiv:2609.15364. (curriculum/actor/verifier three roles, BRS+DRS broad-then-deep memory construction)

UDOS paper volumes (this series, evidence grades marked in text)

1. UDOS v7.5.0 paper volume: P4 "Coverage-aware data flywheel" (coverage-aware active collection), P6 "Observability and identifiability empirical study of hidden dynamics parameters" (acceleration scalar skill about -0.508 , single seed, cpu-proxy), P9 "Agent legion quality saturation law and homogeneous expansion failure boundary" ($mse(k)=a+b/k$, cpu-proto), P10 "TransferBundle handoff information loss measurability."
 2. UDOS v7.6.0 paper volume: P33 "v7.6.0 architecture master: eight structural change specifications and compatibility paths" (BRS+DRS registered as structural change item 3).
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Evidence Discipline and Reproduction Notes

- This paper is an **architecture design paper**, no new experiments; BRS/DRS objective functions, 60/40 initial, τ_{cov} threshold are all design parameters, actual numerical calibration [RESULT NEEDED], forbidden to write expected benefits as measured.
- Citing RSIAgent only takes its BRS/DRS strategy structure; its "open source surpasses closed source" performance conclusion is paper/media report, not independently reviewed, not serving as this paper's argument.
- P6 acceleration scalar skill ≈ -0.508 is a single-seed pilot (cpu-proxy), this paper only points back, not recomputes; before submission must supplement ≥ 30 seeds and 95% CI.
- The two-stage flywheel's causal claims (F18-a/b) are post-hoc-scorable design predictions, not established conclusions.